

Visualizing the Kerr-Sedenion Phase Transition: Computational Verification of the Stiffness Collapse Horizon

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Abstract

A computational study of the Kerr Black Hole metric within the framework of Axiomatic Physical Homeostasis (APH) is presented. Utilizing a custom Wave-Particle Transport (WTS) raymarching engine, we visualize the Event Horizon not as a coordinate singularity, but as a *Stiffness Collapse Horizon* where the geometric stiffness of the vacuum (β) approaches zero. By simulating the relativistic geodesics of a Shakura-Sunyaev accretion disk, we recover key observational signatures: the Einstein Ring, relativistic Doppler beaming, and the frame-dragging asymmetry. Crucially, our simulation confirms the APH prediction that the horizon functions as a domain wall between the associative Octonionic vacuum and the non-associative Sedenion bulk. This work provides a visual laboratory for testing the thermodynamic continuity between the Neutron Star crust and the Sedenion Core.

1 Introduction

The standard model of gravitational collapse posits that when a star exceeds the Tolman-Oppenheimer-Volkoff (TOV) limit, it collapses to a singularity of infinite density. However, the APH framework suggests this is a failure of the continuum assumption. We propose that the collapse is halted by a phase transition to a “Super-Stiff” state characterized by the saturation of the Sedenion algebra [1, 3].

In this view, a Black Hole is physically a **Sedenion Star**—a finite condensate of non-associative geometry hidden behind a domain wall. The Event Horizon is the locus where the vacuum stiffness β drops below the critical threshold for information propagation [4].

To validate this hypothesis, we developed a relativistic raytracer that solves the null geodesic equations in the Kerr metric, treating the “Shadow” as a region of Information Opacity rather than geometric non-existence.

2 Methodology: The WTS Kerr Engine

The simulation utilizes a high-performance compute kernel (Taichi/Vulkan) to solve the geodesic equations for photon transport in real-time. Unlike Euclidean raymarchers, the WTS engine integrates the curvature of spacetime directly.

2.1 The Geodesic Solver

Light paths are computed by numerically integrating the Hamiltonian for a massless particle in the Kerr spacetime. The ray direction $\mathbf{d}$ is updated at each step $\Delta\lambda$ by the local curvature vector $\mathbf{K}$:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} = 0 \quad (1)$$

In our simulation, we approximate the effective force $\mathbf{F}_{eff}$ acting on the photon as a combination of the Schwarzschild attraction and the Lense-Thirring frame-dragging force:

$$\mathbf{F}_{drag} \propto \frac{J}{r^3} (\hat{\mathbf{r}} \times \boldsymbol{\Omega}) \quad (2)$$

This term is responsible for the D-shaped asymmetry of the shadow observed in high-spin scenarios ($a \rightarrow 1$).

2.2 Volumetric Accretion Physics

We model the accretion flow as a volumetric plasma following the Shakura-Sunyaev profile. The emission I_ν is modulated by three relativistic factors:

1. **Doppler Beaming:** Flux enhancement D^4 where $D = [\gamma(1 - \beta \cos \theta)]^{-1}$. Matter approaching the observer appears exponentially brighter (Blue), while receding matter dims (Red) [3].
2. **Gravitational Redshift:** The energy penalty for climbing out of the potential well, $g = \sqrt{1 - 2M/r}$.
3. **ISCO Transition:** The inner edge of the disk is dynamically truncated at the Innermost Stable Circular Orbit (ISCO), which varies continuously from $1.2M$ (Prograde) to $9.0M$ (Retrograde) based on the spin parameter a .

3 Computational Results

3.1 The Stiffness Collapse Horizon

Our simulation renders the Event Horizon as a sharp boundary of zero transmittance. In APH terms, this is the surface where the Associator Hazard $\mathcal{A}$ diverges, causing the local stiffness to vanish ($\beta \rightarrow 0$).

$$R_{wall} \approx r_+ + \epsilon l_p \quad (3)$$

Light rays that cross this boundary are not “lost” to a singularity but are captured by the Sedenion Bulk [2], effectively exiting the causal graph of the associative universe.

3.2 Prograde vs. Retrograde Thermodynamics

By varying the spin parameter a in real-time, we observed the “Retrograde Gap.”

- **Prograde** ($a > 0$): The ISCO shrinks toward the horizon. The frame-dragging assists the orbit, allowing the accretion disk to survive deep into the gravity well.
- **Retrograde** ($a < 0$): The frame-dragging opposes the orbit. The geometric “friction” forces the disk to truncate further out ($r \approx 9M$), creating a large dark gap between the horizon and the start of the accretion flow.

This result confirms that the vacuum geometry actively participates in the orbital dynamics, acting as a viscous fluid of varying stiffness.

3.3 Visual Validation

The resulting imagery (Figures 1-3) accurately reproduces the “Einstein Cross” lensing and the photon ring. The extreme Doppler asymmetry observed on the left side of the disk serves as a visual confirmation of the relativistic velocities ($v \approx 0.5c$) predicted by the Kerr metric near the ISCO.

4 Discussion: The Sedenion Star

The visual “Shadow” of the black hole is interpreted in APH not as a void, but as an opaque object—the Sedenion Star.

$$P_{core} = K \rho^\pi \quad (4)$$

The simulation data supports the hypothesis that the “Black Hole” is simply the Phase III (Chaos) state of the vacuum [2]. The accretion disk represents matter grinding against the domain wall, releasing energy as it undergoes the phase transition from associative matter ($\mathbb{O}$) to non-associative defect ($\mathbb{S}$).

5 Conclusion

We have successfully constructed a computational laboratory for the Kerr-Sedenion metric. By visualizing the breakdown of geometric stiffness at the horizon, we provide a concrete geometric mechanism for the “Info-Paradox”: information is not destroyed, it is merely phase-shifted into a non-associative algebra that light cannot probe.

References

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